

Probability theory I, Fall 2023

Quiz 1

October 17, 2023

Name: _____

Student No.: _____

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2	25	
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5	15	
6	15	
Total	100	

1. (15 points) Suppose there are two coins. Coin 1 lands on heads with probability $\frac{1}{2}$, coin 2 lands on heads with probability $\frac{1}{3}$. We randomly select one coin and repeatedly flip it. If the first flip results in heads, what is the conditional probability that there are exactly 3 heads in the first 5 flips? (Express your answer in the fraction form)

Solution. Let H_1 be the event that the first flip results in head. Let C_1, C_2 be the events that coin 1, 2 are selected, respectively. Let E be the event that there are exactly 3 heads in the first 5 flips.

By law of total probability for conditional probability,

$$\begin{aligned}\mathbb{P}(E|H_1) &= \mathbb{P}(E|H_1C_1)\mathbb{P}(C_1|H_1) + \mathbb{P}(E|H_1C_2)\mathbb{P}(C_2|H_1) \\ &= \binom{4}{2} \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^2 \mathbb{P}(C_1|H_1) + \binom{4}{2} \left(\frac{1}{3}\right)^2 \left(\frac{2}{3}\right)^2 \mathbb{P}(C_2|H_1).\end{aligned}$$

By Bayes's formula,

$$\mathbb{P}(C_1|H_1) = \frac{\mathbb{P}(H_1|C_1)\mathbb{P}(C_1)}{\mathbb{P}(H_1|C_1)\mathbb{P}(C_1) + \mathbb{P}(H_1|C_2)\mathbb{P}(C_2)} = \frac{3}{5},$$

and $\mathbb{P}(C_2|H_1) = 1 - \mathbb{P}(C_1|H_1) = \frac{2}{5}$. Altogether,

$$\mathbb{P}(E|H_1) = \frac{3}{8} \cdot \frac{3}{5} + \frac{24}{81} \cdot \frac{2}{5}.$$

2. Answer the following questions:

- (a) (5 points) How many terms are there in the expansion of $(2x + y + 3z)^5$? What is the coefficient of x^2y^2z ?
- (b) (5 points) 3 married couples are seated in a row, in how many ways can they sit if no person is next to their partner?
- (c) (5 points) E, F are mutually exclusive events with $\mathbb{P}(E) = 0.3, \mathbb{P}(F) = 0.4$. Compute $\mathbb{P}(E|E \cup F)$.
- (d) (5 points) A, B, C are three events. Show that $\mathbb{P}(A \cup B \cup C) \geq \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) - 2$.
- (e) (5 points) Prove or give a counterexample: if E and F are independent, then E and F are conditionally independent.

Solution.

- (a) There are $\binom{5+3-1}{3-1} = 21$ terms and the coefficient of x^2y^2z is

$$\binom{5}{2, 2, 1} 2^2 \cdot 3 = 360.$$

- (b) Let A_i be the event that couple i sit next to each other, then

$$\begin{aligned} \mathbb{P}(\text{no couple sits together}) &= 1 - \sum_{i=1}^3 \mathbb{P}(A_i) + \sum_{i < j} \mathbb{P}(A_i A_j) - \mathbb{P}(A_1 A_2 A_3) \\ &= 1 - 3 \frac{2 \cdot 5!}{6!} + 3 \frac{2^2 \cdot 4!}{6!} - \frac{2^3 \cdot 3!}{6!} \\ &= \frac{1}{3}. \end{aligned}$$

The number of arrangements is $6! \cdot 1/3 = 240$.

- (c) E, F are mutually exclusive, then $\mathbb{P}(E \cup F) = \mathbb{P}(E) + \mathbb{P}(F) = 0.7$ and

$$\mathbb{P}(E|E \cup F) = \frac{\mathbb{P}(E)}{\mathbb{P}(E \cup F)} = \frac{3}{7}.$$

- (d) Since $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(AB)$, we have

$$\begin{aligned} \mathbb{P}(A \cup B \cup C) &= \mathbb{P}(A \cup B) + \mathbb{P}(C) - \mathbb{P}((A \cup B) \cap C) \\ &= \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(AB) - \mathbb{P}((A \cup B) \cap C) \geq \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) - 2. \end{aligned}$$

- (e) Consider the counterexample with $G = E \cup F$, then $\mathbb{P}(E|G) = \frac{\mathbb{P}(E)}{\mathbb{P}(E \cup F)}$, $\mathbb{P}(F|G) = \frac{\mathbb{P}(F)}{\mathbb{P}(E \cup F)}$, but

$$\mathbb{P}(EF|G) = \frac{\mathbb{P}(EF)}{\mathbb{P}(E \cup F)} \neq \mathbb{P}(E|G)\mathbb{P}(F|G) \text{ whenever } 0 < \mathbb{P}(E \cup F) < 1.$$

3. A, B take turns tossing a fair six-sided die in the way that, one stops tossing until getting a 6, A starts the game. That is, A starts tossing and stops when getting a 6, then B tosses until getting a 6, and so on. Let E_n be the event that A tosses at the n^{th} time, and $P_n = \mathbb{P}(E_n)$.

(a) (3 points) Find P_1, P_2 .

(b) (12 points) Find P_n .

Solution.

(a) A tosses first, so $P_1 = 1$. Note that

$$P_2 = \mathbb{P}(A \text{ does not get a 6 in the first toss}) = \frac{5}{6}.$$

(b) We condition on E_{n-1}, E_{n-1}^c ,

$$\begin{aligned} P_n &= \mathbb{P}(E_n) = \mathbb{P}(E_n|E_{n-1})\mathbb{P}(E_{n-1}) + \mathbb{P}(E_n|E_{n-1}^c)\mathbb{P}(E_{n-1}^c) \\ &= P_{n-1}\mathbb{P}(E_n|E_{n-1}) + (1 - P_{n-1})\mathbb{P}(E_n|E_{n-1}^c). \end{aligned}$$

Note that $\mathbb{P}(E_n|E_{n-1}) = \mathbb{P}(A \text{ does not get a 6}) = \frac{5}{6}$, and $\mathbb{P}(E_n|E_{n-1}^c) = \mathbb{P}(B \text{ gets a 6}) = \frac{1}{6}$.

Thus, for $n \geq 2$,

$$\begin{aligned} P_n &= \frac{5}{6}P_{n-1} + \frac{1}{6}(1 - P_{n-1}) = \frac{1}{6} + \frac{2}{3}P_{n-1} = \cdots = \left(\frac{2}{3}\right)^{n-1} P_1 + \sum_{k=0}^{n-2} \frac{1}{6} \left(\frac{2}{3}\right)^k \\ &= \left(\frac{2}{3}\right)^{n-1} + \frac{1}{6} \frac{1 - \left(\frac{2}{3}\right)^{n-1}}{\frac{1}{3}} \\ &= \left(\frac{2}{3}\right)^{n-1} + \frac{1}{2} \left[1 - \left(\frac{2}{3}\right)^{n-1}\right]. \end{aligned}$$

4. 8 keys are identical in form and only one can open the door. You select one key at one time, let X be the number of selections needed to open the door..
- (a) (7 points) Consider put-back case, that is, you put the selected key back after trying it. Give the probability mass function of X and compute the expectation.
- (b) (8 points) Consider no put-back case, that is, you don't put the selected key back after trying it. Give the probability mass function of X and compute the expectation.

Solution.

- (a) X is a geometric random variable with $p = \frac{1}{8}$, hence, the mass function is

$$\mathbb{P}(X = k) = \frac{1}{8} \left(\frac{7}{8}\right)^{k-1} \text{ for } k = 1, 2, \dots$$

Directly, $\mathbb{E}[X] = 8$.

- (b) For the no put-back case,

$$\mathbb{P}(X = k) = \frac{1}{8}, \text{ for } k = 1, 2, \dots, 8.$$

The expectation is

$$\mathbb{E}[X] = \frac{9}{2}.$$

5. (15 points) At all times, a mathematician carries 2 matchboxes: the box in his left-hand pocket contains 2023 matches, the box in his right-hand pocket contains 2024 matches. Each time he needs a match, he takes it from the left pocket with probability $\frac{1}{3}$, from the right pocket with probability $\frac{2}{3}$. Consider the moment when the mathematician first discovers that one of his matchboxes is empty, what is probability that there are exactly k matches in the other box?

Solution. Let E be the event that the mathematician takes a match from the left pocket, and let X be the time he discovers the box in the left pocket is empty. Then X is a negative binomial random variable with $r = 2024, p = \frac{1}{3}$. Similarly, let Y be the first time he discovers the box in the right pocket is empty. Then Y is a negative binomial random variable with $r = 2025, p = \frac{2}{3}$.

The required probability is

$$\begin{aligned} & \mathbb{P}(X = 2024 + (2024 - k)) + \mathbb{P}(Y = 2025 + (2023 - k)) \\ &= \binom{4047 - k}{2023} \left(\frac{1}{3}\right)^{2024} \left(\frac{2}{3}\right)^{2024 - k} + \binom{4047 - k}{2024} \left(\frac{1}{3}\right)^{2023 - k} \left(\frac{2}{3}\right)^{2025} \end{aligned}$$

for $k = 0, 1, \dots, 2023$.

For $k = 2024$, the probability is

$$\mathbb{P}(X = 2024) = \left(\frac{1}{3}\right)^{2024}.$$

6. (15 points) 2023 married couples are randomly seated (all possible orderings being equally likely) at a round table, use Poisson paradigm to approximate the probability that no married couples are seated next to each other.

Solution. Let $C_i, i = 1, 2, \dots, 2023$ represents the event that the i^{th} couple sits together, then

$$\mathbb{P}(C_i) = \frac{2 \cdot 4044!}{4045!} = \frac{2}{4045}.$$

C_i 's are weakly dependent, since

$$\mathbb{P}(C_{i_1} \cdots C_{i_k}) = \frac{2^k \cdot (4045 - k)!}{4045!} \sim \left(\frac{2}{4045} \right)^k = \mathbb{P}(C_{i_1}) \cdots \mathbb{P}(C_{i_k}).$$

Note that the probabilities are small, by Poisson paradigm, the number of these events that occur approximately has a Poisson distribution with mean $\frac{2023 \cdot 2}{4045} = \frac{4046}{4045}$. Thus, the probability that no couple sit next to each other is approximately

$$\exp \left\{ -\frac{4046}{4045} \right\} \approx e^{-1}. \quad \text{both forms are OK here}$$